3 A fixed mass of gas undergoes a cycle of changes of pressure, volume and temperature as shown in Fig. 3.1. The gas is assumed to be ideal.

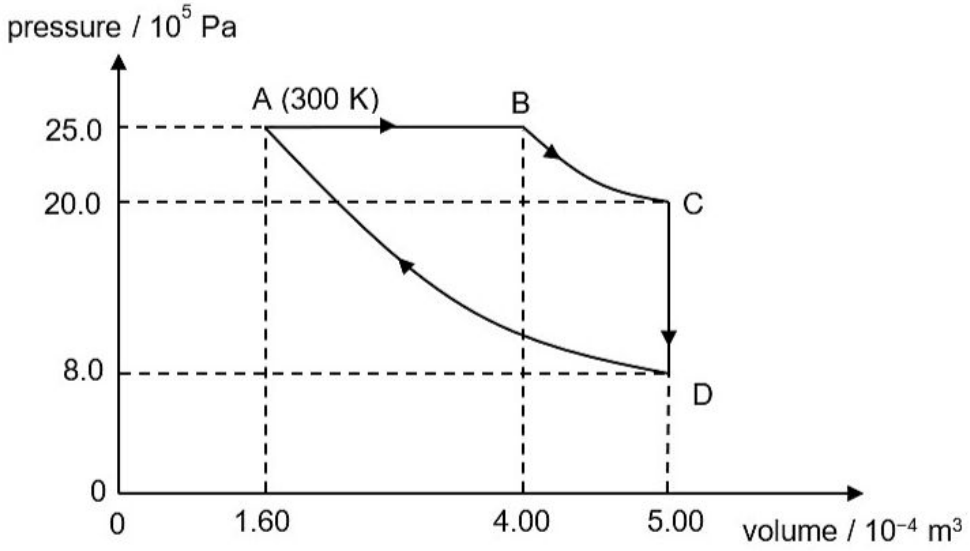


Fig. 3.1 (Not to scale)

The temperature of the gas at A is 300 K.

(a) Show that the amount of gas is 0.160 mol.

$$pV = nRT \Rightarrow n = \frac{pV}{RT} = \frac{(25.0\times10^5)(1.60\times10^{-4})}{(8.31)(300)} \quad \text{[M1]}$$

$$= 0.160 \quad \text{[A0]}$$

[1]

(b) Calculate the temperature at B.

$$pV = nRT \Rightarrow n = \frac{pV}{RT}$$

$$0.16 = \frac{(25.0\times10^5)(4.00\times10^{-4})}{(8.31)(T)} \quad \text{[M1]}$$

$$T = 750 \text{ K} \quad \text{[A0]}$$

temperature = ... K [1]

Fig. 3.2 indicates the type of the various processes in the cycle:

Process	Type of process
A to B	isobaric (constant pressure)
B to C	isothermal (constant temperature)
C to D	isochoric (constant volume)

Fig. 3.2

(c) Explain why process D to A is likely an isothermal process.

from $pV = nRT$. Since n and R are constants, the product pV must remain constant for temperature to be constant for isothermal process. [B1]

Checking the endpoints: at D, $pV = 8.0 \times 10^5 \times 5.00 \times 10^{-4} = 400$ J. At A, $pV = 25.0 \times 10^5 \times 1.60 \times 10^{-4} = 400$ J. [B1]

Since pV is the same at both points D and A, the temperature is unchanged between D and A. Hence the process D to A is likely isothermal.

[2]

Fig. 3.3 lists some data about the gas:

Quantity	Value
molar mass	0.0200 kg mol^{-1}
energy transferred to the gas from A to B by heating	1500 J
specific heat capacity at constant volume	625 J kg^{-1} K^{-1}

Fig. 3.3

(d) (i) Using the data from Fig. 3.3, show that the specific heat capacity of the gas from A to B at constant pressure is 1040 J kg^{-1} K^{-1}.

For process A to B:

$q = mc\Delta T = (0.0200)(0.160)c(750 - 300) = 1500$ J [B1]

$\therefore c = 1040$ J kg^{-1} K^{-1}

[1]

(ii) Use the first law of thermodynamics to determine the increase in internal energy ΔU of the gas from A to B.

$\Delta U = q + w$

$= 1500 + \left[-\left(25.0 \times 10^5\right)\left(2.40 \times 10^{-4}\right)\right]$ [M1]

$= 900$ J [A1]

Do not accept if formula for internal energy was used to determine ΔU due to question requirement.

ΔU = ... J [2]

(iii) Using the data from Fig. 3.3, calculate the heat flow during the process C to D.

process C to D:

$$q = mc\Delta T$$
$$= (0.0200)(0.160)(625)(300 - 750)$$
$$= -900 \text{ J} \quad \text{[A1]}$$

heat flow during the process C to D = ... J [1]

(e) Fig. 3.4 shows the increase in internal energy ΔU of the gas, the heat supplied q to the gas, and the work done on the gas w during the various processes of the cycle. During B to C, 220 J of energy is transferred to the gas by heating and during D to A, work done on the gas is 320 J.

Process	ΔU / J	q / J	w / J
A to B	**900**	**1500**	**−600**
B to C	**0**	**220**	**−220**
C to D	**−900**	**−900**	**0**
D to A	**0**	**−320**	**320**

Fig. 3.4 (1 mark for each correct row.)

Complete the table in Fig. 3.4. [4]

[Total: 12]